AUTOMATED DAILY LITERATURE DIGEST • PhD RESEARCH WATCH

Particulate Matter
Monitoring & Health Effects

Issue 2026-11-04 | Entry window 4 November 2026 (full day) | 18 records screened in scope

18	7	15	3	3
RECORDS IN SCOPE (70 EXCLUDED)	OF 10 SUBTOPIC BANDS POPULATED	EFFECT ESTIMATES WITH A CI	TIER-A RECORDS	RECORDS CARRIED ON METADATA ALONE

Signal of the day

A metrology-chain network calibration worked for temperature, humidity and pressure in Helsinki and Antwerp and failed first on PM: feasible only on restricted sensor subsets, because PM signals are localized and weakly correlated.

VTT-led; Laplace-domain pairwise transfer functions, a separated spatial term and an overdetermined solve against one reference, with Monte Carlo uncertainty. Helsinki linked to SMEAR III; Antwerp from SensEURCity. **Where it is weak:** the abstract gives no PM R^2, RMSE or uncertainty, so the size of the failure cannot be graded, and two cities do not bound the geometries where it holds. **Why it matters here:** node-to-node calibration transfer assumes neighbouring sensors see related signals. The more local the source field, the more value a dense network has and the less that assumption holds; the property that justifies deployment is the one that defeats reference-free transfer. Co-location does not go away.

What changed today

- **Life-course composition, on a 1930–2010 CTM.** UK Biobank, 47,737 with birth-to-baseline residences, 13–14 years of follow-up. Per IQR of $PM_{2.5}$: COPD HR 1.138 (1.069–1.211), asthma 1.171 (1.105–1.241), lower respiratory infection 1.046 (1.002–1.091), lung cancer 1.193 (1.064–1.337); SO_4 and EC positive for all four. IQR not in the abstract; most of the period predates any validating monitor.
- **A joint HR that is not an interaction.** CFPS China, 9,265 adults: $PM_{2.5}$ HR 1.54 (1.42–1.67) per unstated IQR for incident CVD; high $PM_{2.5}$ with compound dry-hot extremes 1.92 (1.67–2.21), called synergistic with no interaction scale given. An IQR estimate that large sits well above the cohort literature; the abstract names neither exposure source nor CVD ascertainment.
- **Tenfold regional PMF in a regulated industry.** NIOSH surveillance, 30,319 underground coal miners, 1991–2024: Central Appalachia PMF PR 10 (5.7–17.5) and CWP PR 3.9 (3.4–4.5), tenure-adjusted; continuous miner operators PMF PR 3.0 (2.3–4.0). Job and region proxy for dust; nothing was measured.
- **Sample count as the exposure instrument.** Colorado State: five people ran 83 AirPen personal monitors across one shift at a plant of about 100 workers; 67 valid PM and noise samples, 22 formaldehyde. Intervals grew several-fold at smaller n; one shift says nothing about day-to-day variance. BACKFILL, entry 28 May.
- **Two sensor evaluations, two standards of evidence.** ARMIE (SPS30, Aarhus): r 0.93–0.98 against a DustTrak but 0.74–0.94 against SMPS+APS, tested leave-one-source-out, the design epidemiology needs. Goyal (two PA-II, one FEM, 150 days): +17.55 % raw bias to 0.13 % NMB, R^2 0.9010, with no stated train/test split. Both BACKFILL; only the first informs transfer.
- **Exposure basis (x1).** Of 18 records, 3 rest on a low-cost sensor or network (network calibration, ARMIE, Goyal) and 1 on a personal monitor (AirPen); none of these is a health study. Health side with a recoverable basis (n=5): 60 % measured, 20 % modelled, 20 % proxy, but that measured share is the AirPen plant survey, six sawmill extracts and a chicken-lung eluate. *No effect estimate below is tied to a measured PM value*: a CTM (Biobank), an unstated surface (CFPS), a pooled mix (rhinitis), job or region (miners).

Window continuity

The 3 November issue closed with `last_entry_date` at 3 November. This issue was built on the **6 November 00:33 UTC (5 November 18:33 CST) run**, a slot that fired 19.5 hours late. Before 22:00 the newest buildable day is yesterday, so **4 November is covered in full and 5 November waits for the next run after 22:00 CST**. Three BACKFILL records, the cap. No rollup is owed: W45 (1–7 November) falls on Saturday 7 November. **The OpenAlex connector was not available on this session**, so the provisional 3 November recovery of that leg remains unconfirmed.

Corpus analytics

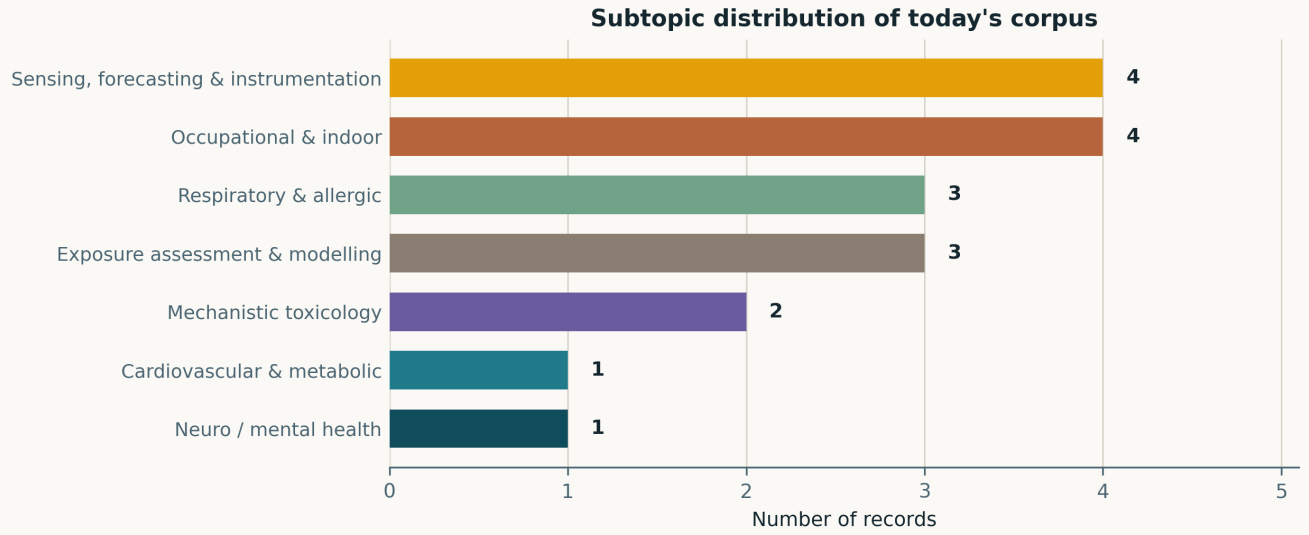


Fig. Subtopic assignment of the 18 records across seven of ten bands: sensing and occupational four each, respiratory and exposure assessment three each, mechanistic two, cardiovascular and neuro one each. Reproductive, burden and other clinical are empty today.

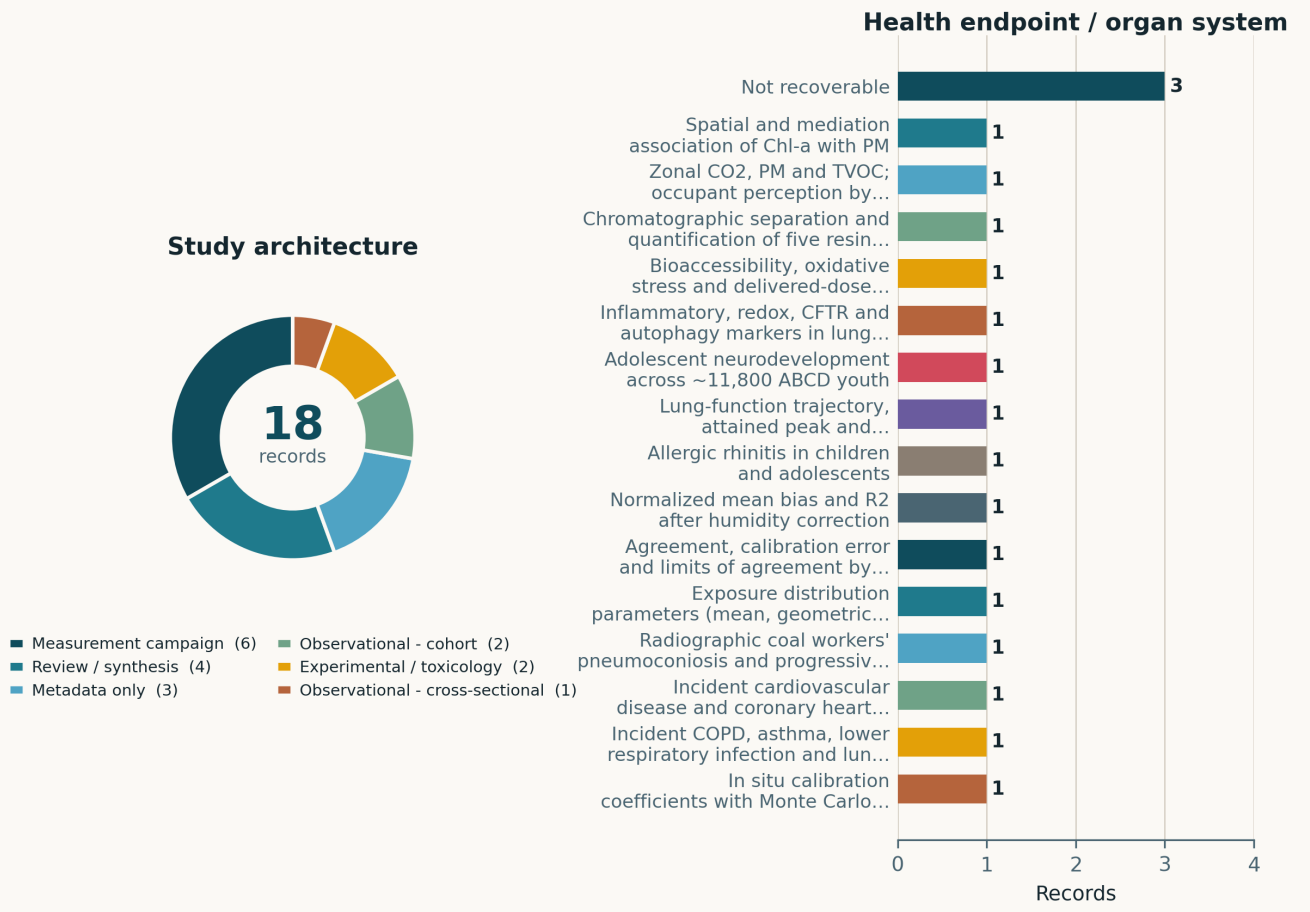


Fig. Left — architecture: measurement campaign six, review or synthesis four, metadata-only three, cohort two, experimental or analytical two, cross-sectional one. Right — endpoints are singletons apart from the three unrecoverable metadata-only records; many are metrological rather than clinical, which is the day's balance.

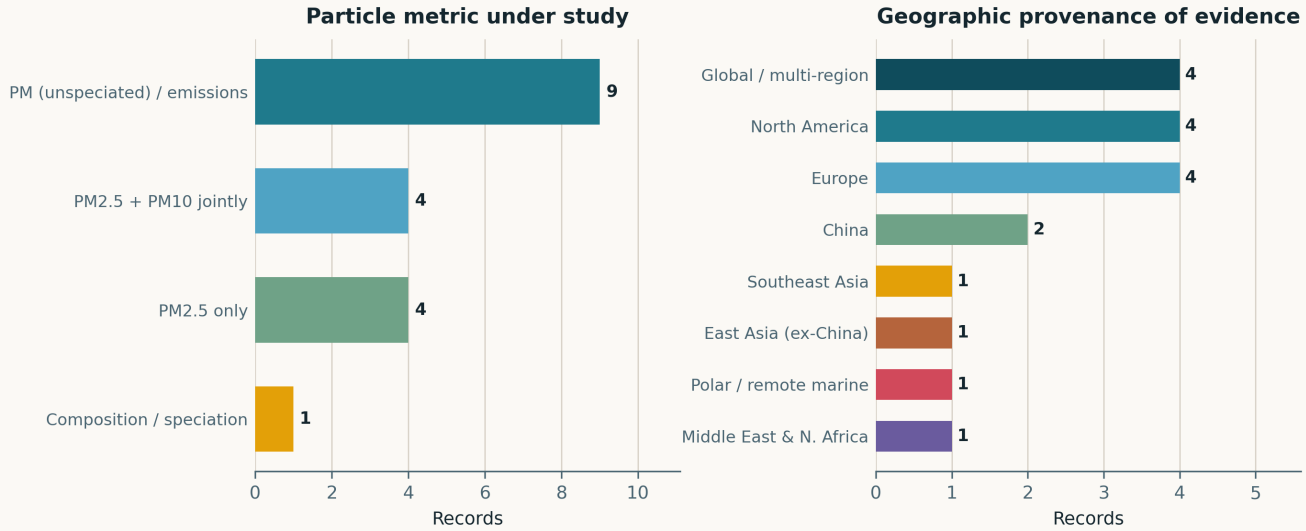


Fig. Left — particle metric: unspeciated or emissions 9, $PM_{2.5}$ only 4, $PM_{2.5}$+PM_{10} jointly 4, composition 1. Right — geography: North America 4, Europe 4 (including the UK), global or multi-region 4 (three reviews, plus one metadata-only record with no location that falls there by default), China 2, and one each Southeast Asia, East Asia ex-China, Middle East and N. Africa (Istanbul) and polar (Ardley Island).

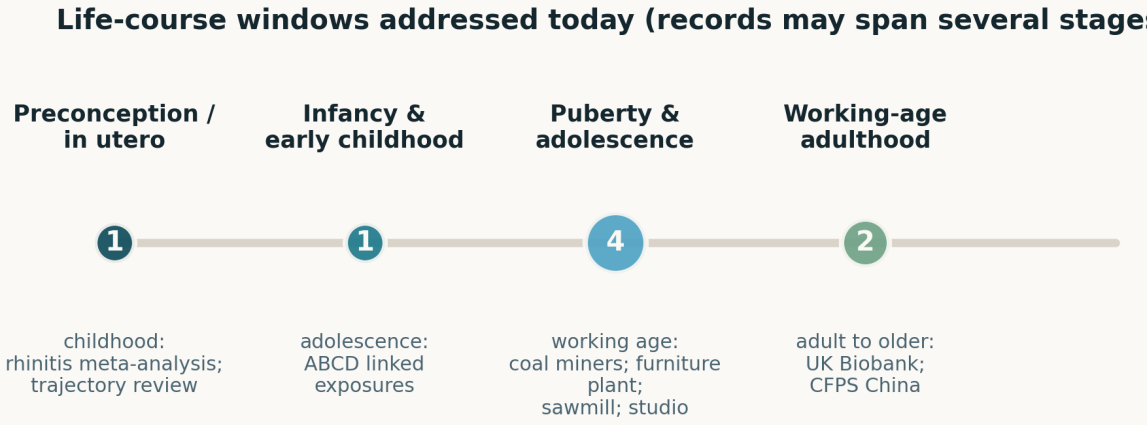


Fig. Life-course windows: childhood (rhinitis meta-analysis, trajectory review), adolescence (ABCD), working age (four occupational and indoor settings), and adult to older (UK Biobank, CFPS). No preconception or in-utero record today.

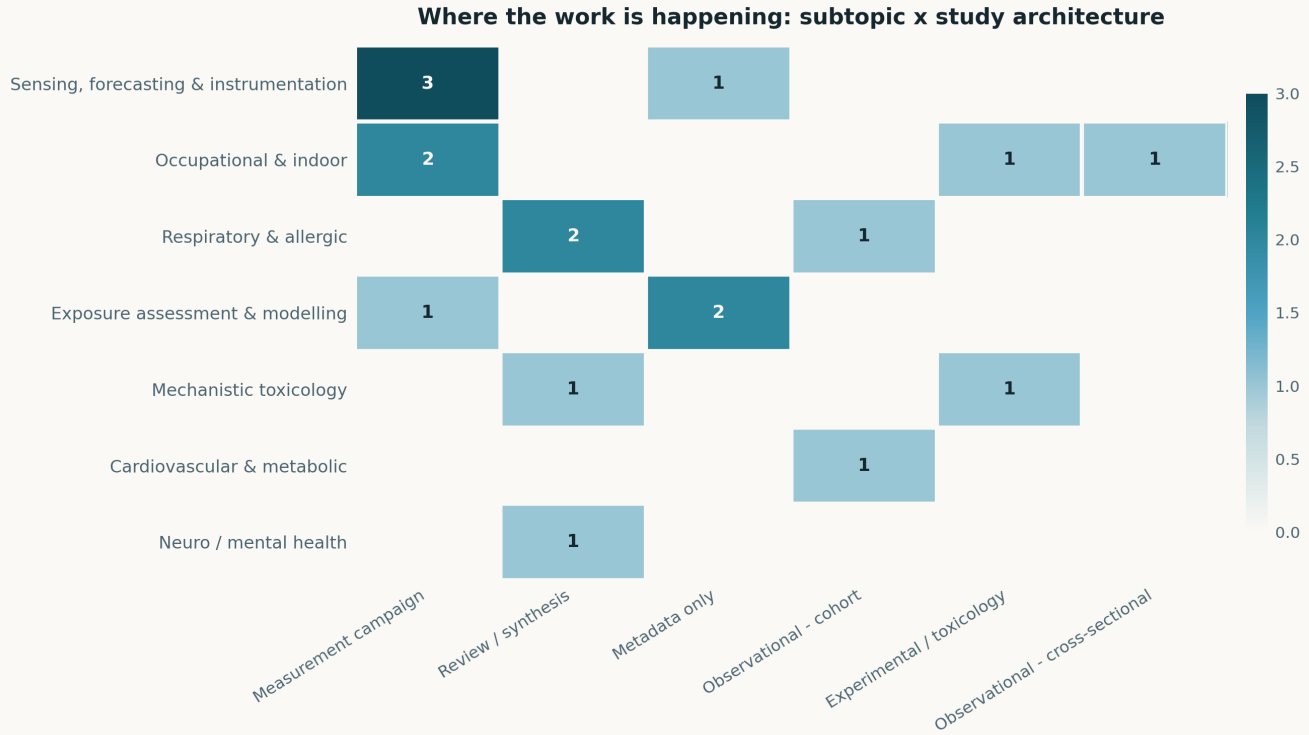


Fig. Subtopic × study architecture. Sensing sits in measurement (3) plus one metadata-only; occupational spreads over measurement (2), analytical chemistry and the miner surveillance record. The only cohorts are respiratory and cardiovascular, and neither carries a measured exposure; no measured-exposure cohort is in the matrix today.

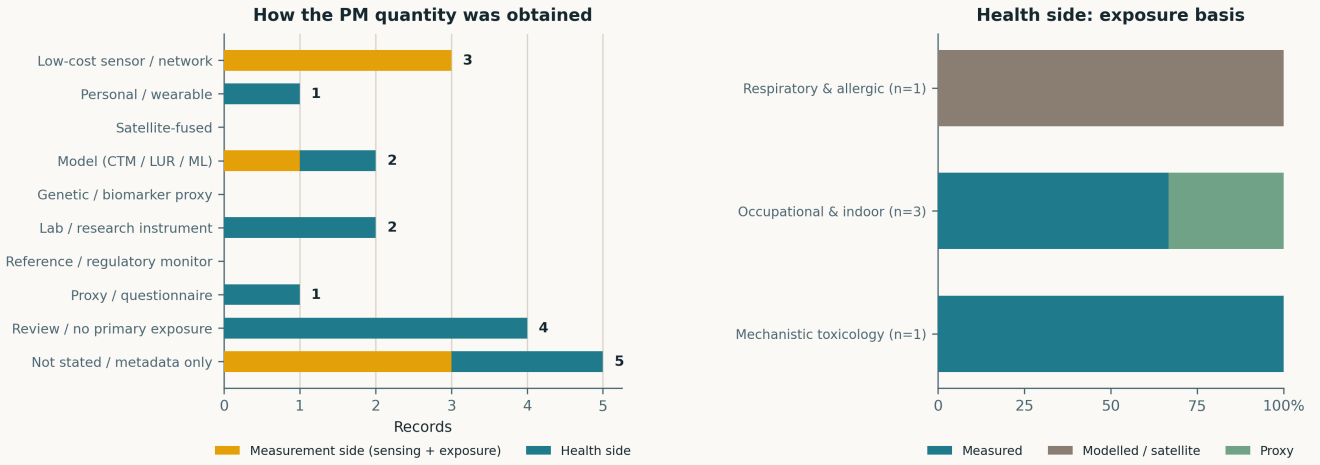


Fig. How each record obtained its PM quantity, classified from the exposure field. 3 of 18 records rest on a low-cost sensor or sensor network; 0 on reference or regulatory monitors. Health side, where an exposure basis is recoverable (n=5): measured 60 %, modelled or satellite-fused 20 %, proxy 20 %. Right panel sorted by modelled share: the subtopics at the bottom are where a measured low-cost network exposure would add most.

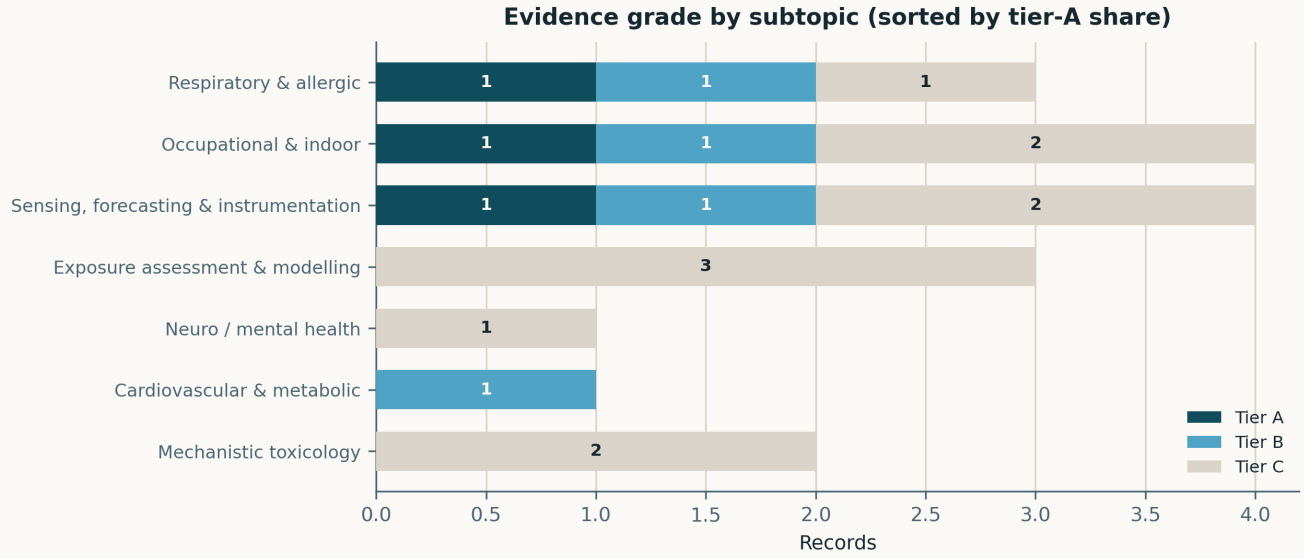


Fig. Tier mix per subtopic. 3 of 18 records are tier A (17 %). Highest tier-A share: *Respiratory & allergic* (1 of 3).

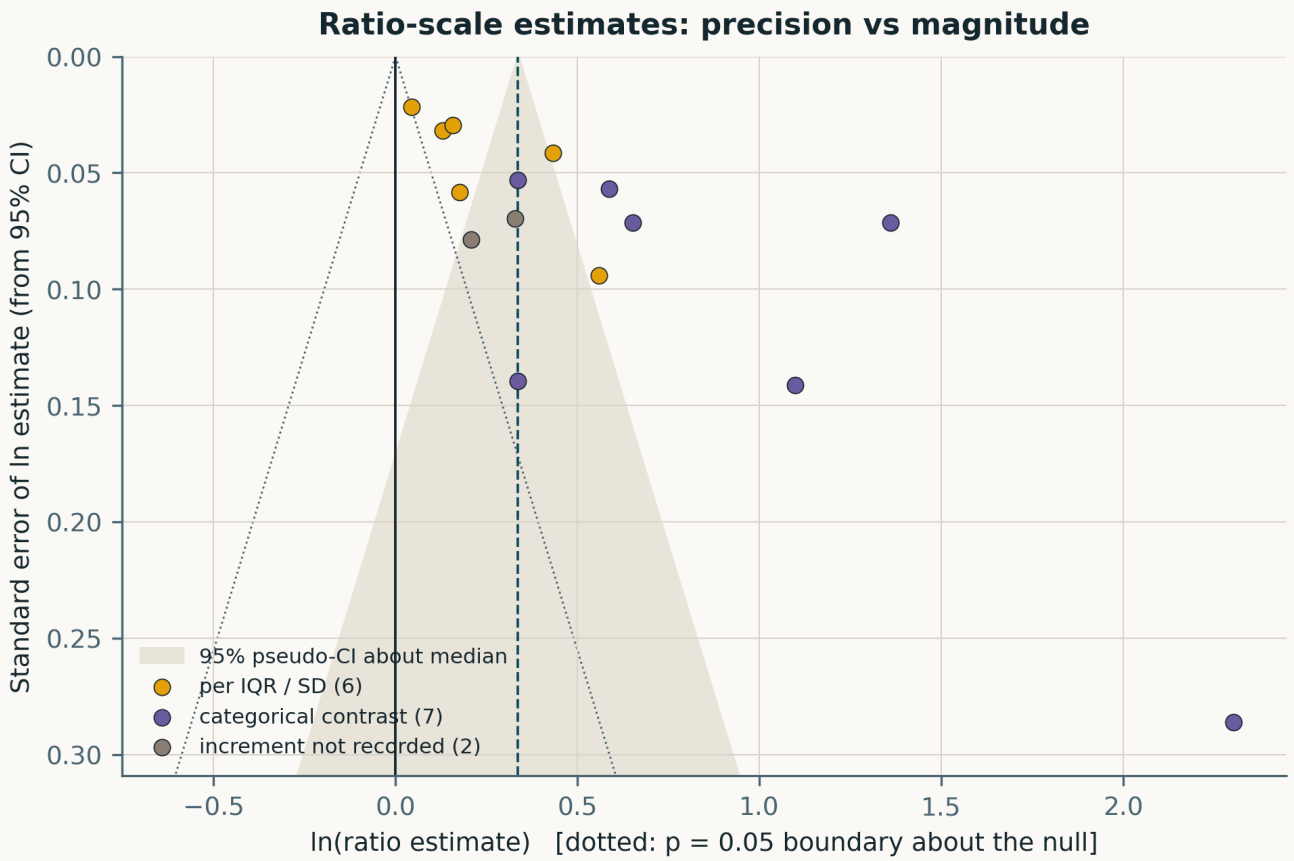


Fig. Every ratio-scale estimate (n=15) plotted as ln(estimate) against its standard error, coloured by the exposure increment the authors reported. 15 of 15 exclude the null. Increments: categorical contrast 7, per IQR / SD 6, increment not recorded 2. Because increments differ (per 1 unit, per 10 $\mu g/m^3$, per IQR, categorical), the spread is not a heterogeneity estimate and the median line is not a pooled effect; the plot shows whether precision and magnitude co-vary. Egger regression not computed: the 15 estimates come from 4 sources and are not independent.

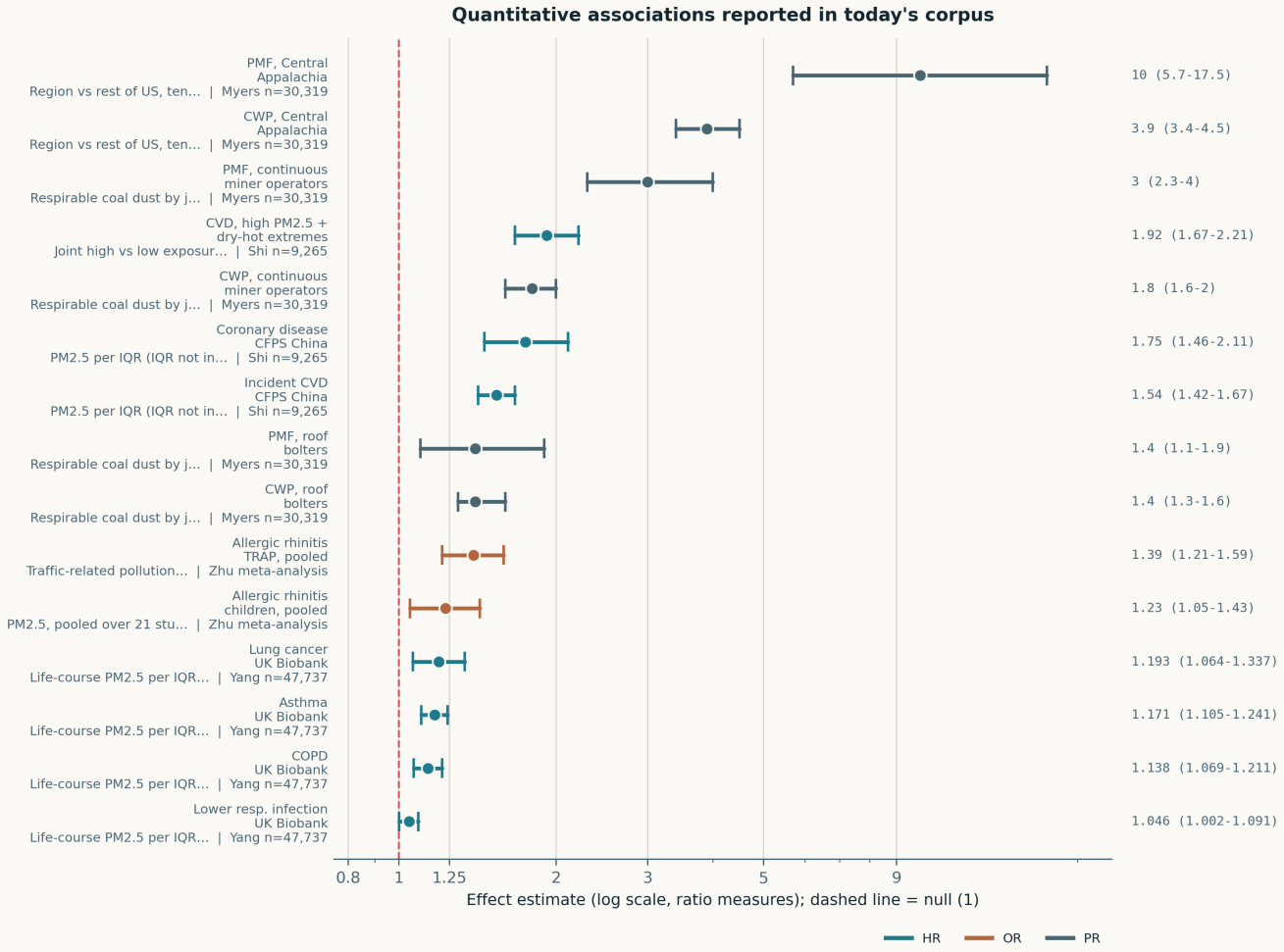


Fig. Ratio scale, 15 estimates from 4 sources: UK Biobank (4), CFPS (3), one meta-analysis (2), NIOSH miners (6). The rhinitis pair carries no increment in its abstract and converts to no concentration contrast. Estimates within a source share a cohort and are not independent. Increments differ (per unstated IQR, categorical contrast, none in the abstract) and PR on prevalence is not HR on incidence, so nothing here is poolable; no I^2 is reported. Upper rows are job and region contrasts in respirable coal dust, not ambient PM.

Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase. Records marked *metadata only* carry no recoverable abstract and assert no finding.

Neuro / mental health — 1 record

Cardenas-Iniguez ABCD linked external data environment review first decade 2026 Dev Cogn Neurosci

Environmental influences using linked external data in ABCD: A review of the first decade.

Ten years of Linked External Data work in the Adolescent Brain Cognitive Development Study (~11,800 US youth), covering neighbourhood conditions, built environment, air pollution, greenspace and state policy. The authors conclude that development reflects cumulative influences and that timing and study design bound what can be concluded, and they call for refined geocoding and exposure methods. No effect sizes in the abstract. Limitation: an internal working-group review of its own dataset, and residential geocoding ignores school and transit time, which matters at this age. Relevance: moderate - the largest US neurodevelopment cohort names exposure resolution as a limiting method.

doi: 10.1016/j.dcn.2026.101822

Cardiovascular & metabolic — 1 record

Shi CFPS compound dry-hot extremes $PM_{2.5}$ joint incident CVD China 2026 BMC Public Health

Joint effects of air pollution and compound dry-hot extremes on cardiovascular disease risk across adults: a cohort study in China.

China Family Panel Studies, 9,265 adults with 69,297 repeated observations, 2010-2018, mixed-effects Cox. Per IQR, $PM_{2.5}$ HR 1.54 (1.42-1.67) for CVD and 1.75 (1.46-2.11) for coronary disease; dry-hot cumulative days HR 1.30 (1.20-1.41). Concurrent high exposure to both gave a joint HR of 1.92 (1.67-2.21), called synergistic, and repeated across PM_1, PM_{10}, NO_2 and O_3. Limitation: an IQR HR of 1.54 is far above the cohort literature and neither the IQR nor the exposure source is stated; the abstract does not specify how CVD was ascertained; joint exposure and synergy are different quantities, and no interaction scale is given. Relevance: low - no measurement content, but a compound-exposure design that a network could sharpen.

doi: 10.1186/s12889-026-28927-4

Respiratory & allergic — 3 records

Yang UK Biobank life-course $PM_{2.5}$ constituents COPD asthma LRTI lung cancer 2026 Lung

Life-Course Exposure to PM(2.5) Constituents and Major Respiratory Diseases: Constituent-Specific Associations and Candidate Immune-Metabolic Signatures in a Large Prospective Cohort.

47,737 UK Biobank participants with birth-to-baseline residential histories, 13-14 years of follow-up. Per IQR of life-course $PM_{2.5}$: COPD HR 1.138 (1.069-1.211), asthma 1.171 (1.105-1.241), lower respiratory infection 1.046 (1.002-1.091), lung cancer 1.193 (1.064-1.337). SO4 and EC were positive for all four outcomes; mixture associations held for COPD and asthma. The IQR in $\mu g\,m^{-3}$ is not given in the abstract. Limitation: decades of exposure rest on one retrospective CTM with no available monitor validation for most of the period, and the 47,737 with histories are a selected fraction of the Biobank. Relevance: moderate - composition, not mass, is the signal, which a mass-calibrated optical sensor cannot resolve.

doi: 10.1007/s00408-026-00948-1

Zhu meta-analysis air pollutants allergic rhinitis children $PM_{2.5}$ TRAP mold 2026 Int J Pediatr Otorhinolaryngol

Association between exposure to gaseous air pollutants and the risk of allergic rhinitis in children and adolescents: A meta-analysis.

Meta-analysis of 21 studies (13 cross-sectional, 6 cohort, 2 case-control). Pooled ORs: mold 1.45 (1.16-1.80), traffic-related pollution 1.39 (1.21-1.59), $PM_{2.5}$ 1.23 (1.05-1.43), O_3 1.15 (1.05-1.26), NO_2 1.09 (1.00-1.17); PM_{10}, SO_2 and CO not significant. Estimates for PM_{10}, NO_2 and O_3 were higher in Chinese studies. Limitation: the abstract gives no exposure increment, so the pooled ratios cannot be compared across pollutants or to any concentration, and 13 of 21 studies are cross-sectional. Mold is not an ambient pollutant, and a 'gaseous' title over a $PM_{2.5}$ estimate shows loose scoping. Relevance: low - a pooled OR with no increment cannot feed a sensor-based exposure-response.

doi: 10.1016/j.ijporl.2026.113068

Drummond mini-review climate exposures paediatric lung-function trajectories 2026 Eur Respir Rev

Climate change as a modifier of paediatric lung-function trajectories.

Mini-review using lung-function trajectories as its frame. Climate-sensitive exposures shift trajectories downward by three routes: early developmental disruption (gestational pollution and temperature, via growth restriction and preterm birth), sustained exposure impairing postnatal growth and peak, and exacerbations with incomplete recovery. Evidence that cleaner air accelerates lung-function growth is taken as proof the trajectory can be protected. No effect sizes or search method in the abstract. Limitation: co-occurring heat, smoke, allergen and damp are not separated, so attribution to PM is not possible from this frame. Relevance: low - frames why childhood exposure windows matter, without a measurement requirement.

doi: 10.1183/16000617.0222-2026

Mechanistic toxicology — 2 records

Li poultry-house $PM_{2.5}$-derived material APEC sequential exposure chick cell 2026 Poult Sci

Inflammatory and redox responses associated with sequential exposure to poultry-house $PM_{2.5}$-derived material and avian pathogenic Escherichia coli in chicken models.

Chicken type II-like lung cells and 40 SPF chicks (n=10/group, 14 days) received control, single or sequential $PM_{2.5}$-derived material then avian pathogenic E. coli. The sequential group showed the largest changes in several inflammatory and redox endpoints and lower CFTR; N-acetyl-L-cysteine and CBIQ shifted cell responses toward control. No effect sizes in the abstract. Limitation, stated by the authors: the eluted, lyophilized, autoclaved material was not characterized for size, composition or endotoxin, and schedules differed in timing, so no interaction, synergy or particle-specific effect is established. Relevance: low - a sterilized eluate is not an airborne particle.

doi: 10.1016/j.psj.2026.108064

Qureshi narrative review bioaccessibility particle-bound metals PAHs tiered framework 2026 Research Square (preprint)

Bioaccessibility and Toxicity of Particle-Bound Metals and Polycyclic Aromatic Hydrocarbons: Sources, Respiratory Fate, Mechanisms, and Risk Assessment

Single-author critical narrative review, not peer-reviewed. Metal release varies with element, source, size, extraction conditions and solid-to-liquid ratio, and acidic lysosomal fluid usually dissolves more than near-neutral extracellular fluid; PAH release depends on hydrophobicity, soot binding, surfactants and proteins. Poorly soluble particles can stay active with little released. Proposes a tiered framework joining mass, source, size, speciation, compartment-specific bioaccessibility, oxidative potential and delivered dose. No search method or quantitative synthesis in the abstract, and differences in simulated lung fluids limit comparison. Relevance: low - explains why equal mass need not mean equal hazard, beyond what a mass sensor reports.

doi: 10.21203/rs.3.rs-11258579/v1

Exposure assessment & modelling 3 records

Wu penguin colony chlorophyll-a mediation PM_1 $PM_{2.5}$ PM_{10} Ardley Island 2026 Mar Pollut Bull

Phytoplankton-mediated coupling between penguin colonies and atmospheric particulate matter in Antarctic coastal waters.

Synchronous ocean-atmosphere transects around Ardley Island. After controlling for spatial autocorrelation, chlorophyll-a remained associated with $PM_{2.5}$ (standardized coefficient 0.12). Across PM_1, $PM_{2.5}$ and PM_{10} indirect NH4-Chl-a-PM effects exceeded direct ones, up to eightfold, but only at alpha 0.10, with 95% CIs including zero. The authors nonetheless propose Chl-a as a proxy for the colony atmospheric footprint. Limitation: the mediation result does not exclude zero, the transects are one spatial sample, and particle composition is not reported, so marine biogenic origin is inferred. Relevance: low - a very clean background where optical PM readings are near any low-cost sensor's noise floor.

doi: 10.1016/j.marpolbul.2026.120512

Kim CMAQ-UNIPAR secondary organic aerosol South Korea ASIA-AQ 2026 Atmos Environ

Prediction of Secondary Organic Aerosol Formation in South Korea using CMAQ-UNIPAR during the ASIA-AQ Campaign

METADATA ONLY. Metadata only; no abstract in Crossref at issue time, so no finding is asserted. The title places a CMAQ run with the UNIPAR SOA module against the ASIA-AQ campaign over South Korea. Carried because a mechanistic SOA module tested against an airborne and ground campaign bears on the organic fraction of $PM_{2.5}$, the component that most changes refractive index and hygroscopicity across a sensor's calibration range, and on the CTM surfaces health cohorts inherit. To be assessed when the Elsevier abstract deposits: bias against campaign OA and any change in simulated $PM_{2.5}$. Relevance: possible, unassessed.

doi: 10.1016/j.atmosenv.2026.122465

Gantina remote sensing CO HC NOx soot heavy-duty trucks buses Indonesia 2026 Aerosol Air Qual Res

Emission of CO, HC, NOx, and Soot from HDV Trucks and Buses using Remote Sensing in Indonesia

METADATA ONLY. Metadata only; no abstract in Crossref at issue time, so no finding is asserted. The title reports on-road remote sensing of CO, HC, NOx and soot from heavy-duty trucks and buses in Indonesia. Carried because measured real-world fleet soot factors are scarce for Southeast Asia, and because they anchor inventory inputs where reference networks are thin and low-cost networks are the main expansion route. To be assessed when an abstract deposits: fleet size sampled, soot retrieval method and any emission-standard contrast. Relevance: possible, unassessed.

doi: 10.1007/s44408-026-00174-7

Sensing, forecasting & instrumentation 4 records

Soderblom Laplace-domain in situ network calibration Helsinki Antwerp SensEURCity 2026 Meas Sci Technol

Laplace domain dynamic calibration in city-wide sensor networks

A Laplace-domain method estimates pairwise dynamic relationships between sensor signals, separates a medium-related spatial effect, and solves an overdetermined calibration against one known network reference, with Monte Carlo uncertainty. Tested on Helsinki (linked to SMEAR III) and Antwerp (SensEURCity). Coefficients were obtained for temperature, RH and pressure in both networks, but PM calibration was feasible only on restricted sensor subsets because PM signals are localized and weakly correlated. Injected shifts could be flagged when geometry and correlations sufficed. No PM R2 or RMSE in the abstract, so the size of the PM failure cannot be graded. Relevance: direct - a metrology-chain calibration transfer fails first on the measurand networks are built for.

doi: 10.1088/1361-6501/aebdb6

Kloppenborg ARMIE SPS30 node laboratory DustTrak SMPS APS source-out calibration 2026 Sensors

Laboratory Evaluation of ARMIE, a Portable SPS30-Based Low-Cost Sensor Node for $PM_{2.5}$ Monitoring

SPS30 nodes were co-located in the laboratory with a DustTrak and an SMPS+APS across candle, cooking, cigarette smoke and clean air, at regular and high humidity. Agreement with the DustTrak was r 0.93-0.98; with SMPS+APS it was moderate, r 0.74-0.94, with systematic underestimation at high concentration. Leave-one-sensor-out and leave-one-source-out calibration reduced mean error and narrowed limits of agreement. Limitation: a DustTrak is itself a photometer, so the high r partly reflects shared optical bias; the sizing reference is the honest comparator and shows the gap. No gravimetric or field arm, and r is not accuracy. Relevance: direct - leave-one-source-out is the right test for epidemiology nodes. BACKFILL, Crossref 2 January 2026.

doi: 10.3390/s26010280

Goyal PurpleAir PA-II kappa-Kohler Lorentz-Lorenz random forest moisture artefact 2026 Discover Electronics

A physics-aware digital twin for correcting moisture-induced material artifacts in $PM_{2.5}$ sensors

A pair of PA-II units beside one EPA FEM monitor for 150 days in an industrial area. kappa-Kohler growth and a Lorentz-Lorenz refractive index mixing rule motivate features for a polynomial-enhanced random forest. At RH 76.16% optical intensity rose 59.5%, summarised as a mass-intensity divergence factor of 4.33. Raw bias of +17.55% fell to 0.13% NMB, R2 0.9010. Limitation: two sensors at one site, single author, and the abstract gives no train/test separation by time, sensor or site, so 0.13% cannot be read as field skill. Calling FEM 'gravimetric' is loose. Relevance: moderate - the physics is standard and testable; the transferability is untested. BACKFILL, Crossref 26 September 2026.

doi: 10.1007/s44291-026-00286-9

Mao wavelet attention LSTM indoor aerosol concentration prediction 2026 J Aerosol Sci

A Wavelet-Based Attention LSTM Framework for Aerosol Concentration Prediction in Indoor Environments

METADATA ONLY. Metadata only; no abstract in Crossref at issue time, so no finding is asserted. The title describes a wavelet decomposition feeding an attention LSTM to predict indoor aerosol concentration. Carried because short-horizon indoor prediction is one of the candidate edge-inference tasks for a sensor node, and J Aerosol Sci is one of the venues where such work is tied to a measured aerosol. What decides its value is unknown until the abstract deposits: the measuring instrument, whether validation is by held-out episode rather than random split, and the prediction horizon. Relevance: possible, unassessed.

doi: 10.1016/j.jaerosci.2026.106908

Occupational & indoor 4 records

Myers NIOSH CWHSP pneumoconiosis dusty job type underground coal 1991-2024 Front Public Health

Assessment of pneumoconiosis by dusty job type among underground coal miners participating in the Coal Workers' Health Surveillance Program,1991-2024.

30,319 underground miners in the national surveillance programme, 1991-2024, tenure-adjusted. Continuous miner operators had CWP PR 1.8 (1.6-2.0) and PMF PR 3.0 (2.3-4.0) against those never in the job; roof bolters 1.4 (1.3-1.6) and 1.4 (1.1-1.9). CWP prevalence was 8.4% in Central Appalachia versus 1.5% elsewhere, adjusted PR 3.9 (3.4-4.5); PMF PR 10 (5.7-17.5). Limitation: job title stands in for dust, and programme participants are self-selected, so healthy-worker and participation bias run in unknown directions. Prevalence pools 33 years of changing limits. Relevance: moderate - a tenfold regional contrast in a monitored regulated industry is the strongest argument today for personal rather than compliance-point dust data.

doi: 10.3389/fpubh.2026.1970741

Molina Rueda AirPen facility-wide personal PM noise formaldehyde one shift 2026 J Expo Sci Environ Epidemiol

Facility-wide assessment of occupational exposure to dust, noise, and formaldehyde

Five people deployed 83 AirPen compact personal monitors across one full shift at a furniture plant of about 100 workers. Valid after screening: 67 PM, 67 noise, 22 formaldehyde (72, 67, 31 counting samples below the LOD). On-board sensors identified anomalies and confirmed wear. Distribution parameters were estimated with high certainty, and confidence intervals grew several-fold when fewer samples were analysed. Limitation: one facility on one shift says nothing about day-to-day variance, usually the main occupational term; 16 of 83 PM samples were lost or below LOD, and no comparison values are in the abstract. Relevance: direct - sample count, not instrument class, sets the uncertainty. BACKFILL, entry 28 May 2026.

doi: 10.1038/s41370-026-00883-6

Olsen LC-APCI-MS resin acids personal softwood sawmill dust Norway 2026 J Sep Sci

A Versatile Method for Determination of Resin Acids in Occupational Softwood Sawmill Dust Samples by Liquid Chromatography-Atmospheric Pressure Chemical Ionization-Mass Spectrometry.

Six reversed-phase columns and multiple mobile phases were screened; a C18-amide column with acetonitrile and 30 mM ammonium acetate at pH 6.30 and negative APCI gave a 15-minute isocratic run. Applied to six personal sawmill dust extracts, it determined abietic, dehydroabietic and levopimaric acids, coeluting isopimaric and pimaric acids, and 7-oxodehydroabietic acid. Limitation: six extracts demonstrate applicability, not exposure levels, and two isomers still coelute despite that being the problem the method targets. LODs are not in the abstract. Relevance: low - constituent chemistry of an occupational dust that no optical sensor could separate, useful only as a reference analysis.

doi: 10.1002/jssc.70568

Apak office-to-education studio zone CO_2 $PM_{2.5}$ PM_{10} TVOC perception Istanbul 2026 Indoor Air

Indoor Air Quality and Occupant Perception in an Office Building Adapted for Educational Use: A Zone-Based Case Study

One mechanically ventilated design studio, three zones, questionnaire from 48 users. CO_2 was highest and most variable in the central zone (527 +/- 114 ppm, 95th percentile 760 ppm), and upper-end particulate indicators were greater in the circulation zone, but no zonal difference in CO_2, PM or TVOC survived multiplicity adjustment. Perceived freshness tracked overall satisfaction (Spearman rho 0.475, 0.219-0.682). Limitation, stated by the authors: respondents cannot be linked to measurement times or locations, so this is no exposure-perception relationship; sequential monitoring and few points restrict inference. Relevance: low - sequential single-instrument sampling cannot resolve within-room PM differences, which simultaneous nodes could.

doi: 10.1155/ina/4355214

Corpus register

First author	Focus	Journal	Design	Metric	Region
Neuro / mental health					
Cardenas-Iniguez ABCD linked external data environment review first decade	Environmental influences using linked external data in ABCD: A review of the first decade.	*Dev Cogn Neurosci*	Narrative review	Geocoded residential air pollution among linked neighbourhood and policy variables; metric not specified	North America
Cardiovascular & metabolic					
Shi CFPS compound dry-hot extremes $PM_{2.5}$ joint incident CVD China	Joint effects of air pollution and compound dry-hot extremes on cardiovascular disease risk across adults: a cohort study in China.	*BMC Public Health*	Prospective cohort	$PM_{2.5}$ with PM1, PM_{10}, NO2 and O3; exposure source not stated in the abstract	China
Respiratory & allergic					
Drummond mini-review climate exposures paediatric lung-function trajectories	Climate change as a modifier of paediatric lung-function trajectories.	*Eur Respir Rev*	Narrative review	Air pollution and wildfire smoke among climate-sensitive exposures; no PM metric specified	Global
Yang UK Biobank life-course $PM_{2.5}$ constituents COPD asthma LRTI lung cancer	Life-Course Exposure to PM(2.5) Constituents and Major Respiratory Diseases: Constituent-Specific Associations and Candidate Immune-Metabolic Signatures in a Large Prospective Cohort.	*Lung*	Prospective cohort	$PM_{2.5}$, SO4, NO3, NH4, OC and EC from CMAQ 1930-2010 linked to birth-to-baseline residential histories	UK
Zhu meta-analysis air pollutants allergic rhinitis children $PM_{2.5}$ TRAP mold	Association between exposure to gaseous air pollutants and the risk of allergic rhinitis in children and adolescents: A meta-analysis.	*Int J Pediatr Otorhinolaryngol*	Comprehensive review	$PM_{2.5}$, PM_{10}, traffic-related pollution, NO2, O3, SO2, CO and mold as reported by included studies	Global
Mechanistic toxicology					
Li poultry-house $PM_{2.5}$-derived material APEC sequential exposure chick cell	Inflammatory and redox responses associated with sequential exposure to poultry-house $PM_{2.5}$-derived material and avian pathogenic Escherichia coli in chicken models.	*Poult Sci*	Animal + in vitro	Autoclaved eluate of broiler-house $PM_{2.5}$-fraction material, uncharacterized after sterilization	China
Qureshi narrative review bioaccessibility particle-bound metals PAHs tiered framework	Bioaccessibility and Toxicity of Particle-Bound Metals and Polycyclic Aromatic Hydrocarbons: Sources, Respiratory Fate, Mechanisms, and Risk Assessment	*Research Square (preprint)*	Narrative / critical review	Particle-bound metals and PAHs; PM mass, size, speciation and oxidative potential	Global
Occupational & indoor					
Apak office-to-education studio zone CO2 $PM_{2.5}$ PM_{10} TVOC perception Istanbul	Indoor Air Quality and Occupant Perception in an Office Building Adapted for Educational Use: A Zone-Based Case Study	*Indoor Air*	Field measurement	Indoor $PM_{2.5}$ and PM_{10} by sequential point measurement; instrument not named in the abstract	Turkiye
Molina Rueda AirPen facility-wide personal PM noise formaldehyde one shift	Facility-wide assessment of occupational exposure to dust, noise, and formaldehyde	*J Expo Sci Environ Epidemiol*	Field measurement	Total particulate by personal AirPen sampler with on-board sensors, plus formaldehyde and A-weighted noise	North America
Myers NIOSH CWHSP pneumoconiosis dusty job type underground coal 1991-2024	Assessment of pneumoconiosis by dusty job type among underground coal miners participating in the Coal Workers' Health Surveillance Program,1991-2024.	*Front Public Health*	Cross-sectional (occupational)	Respirable coal mine dust, proxied by occupation and region; no dust measurement in the analysis	North America

First author	Focus	Journal	Design	Metric	Region
Olsen LC-APCI-MS resin acids personal softwood sawmill dust Norway	A Versatile Method for Determination of Resin Acids in Occupational Softwood Sawmill Dust Samples by Liquid Chromatography-Atmospheric Pressure Chemical Ionization-Mass Spectrometry.	*J Sep Sci*	Analytical method validation	Resin acids in personal wood dust samples	Europe
Sensing, forecasting & instrumentation					
Goyal PurpleAir PA-II kappa-Kohler Lorentz-Lorenz random forest moisture artefact	A physics-aware digital twin for correcting moisture-induced material artifacts in $PM_{2.5}$ sensors	*Discover Electronics*	Calibration model comparison	$PM_{2.5}$ from a pair of PurpleAir PA-II units co-located with an EPA FEM monitor (site 220330009)	North America
Kloppenborg ARMIE SPS30 node laboratory DustTrak SMPS APS source-out calibration	Laboratory Evaluation of ARMIE, a Portable SPS30-Based Low-Cost Sensor Node for $PM_{2.5}$ Monitoring	*Sensors*	Instrument evaluation	$PM_{2.5}$ from Sensirion SPS30 in a portable node, against DustTrak and combined SMPS+APS	Europe
Mao wavelet attention LSTM indoor aerosol concentration prediction	A Wavelet-Based Attention LSTM Framework for Aerosol Concentration Prediction in Indoor Environments	*J Aerosol Sci*	Metadata only: no abstract in Crossref at issue time	Indoor aerosol concentration	Not stated
Soderblom Laplace-domain in situ network calibration Helsinki Antwerp SensEURCity	Laplace domain dynamic calibration in city-wide sensor networks	*Meas Sci Technol*	Calibration model comparison	PM from deployed urban low-cost sensor networks, alongside temperature, RH and pressure	Europe
Exposure assessment & modelling					
Gantina remote sensing CO HC NOx soot heavy-duty trucks buses Indonesia	Emission of CO, HC, NOx, and Soot from HDV Trucks and Buses using Remote Sensing in Indonesia	*Aerosol Air Qual Res*	Metadata only: no abstract in Crossref at issue time	Soot from on-road heavy-duty vehicles by remote sensing	Indonesia
Kim CMAQ-UNIPAR secondary organic aerosol South Korea ASIA-AQ	Prediction of Secondary Organic Aerosol Formation in South Korea using CMAQ-UNIPAR during the ASIA-AQ Campaign	*Atmos Environ*	Metadata only: no abstract in Crossref at issue time	Secondary organic aerosol	South Korea
Wu penguin colony chlorophyll-a mediation PM1 $PM_{2.5}$ PM_{10} Ardley Island	Phytoplankton-mediated coupling between penguin colonies and atmospheric particulate matter in Antarctic coastal waters.	*Mar Pollut Bull*	Field campaign / shipborne	Along-track PM1, $PM_{2.5}$ and PM_{10} with seawater chlorophyll-a and nutrients; instrument not named in the abstract	Antarctica

Provenance and method

Entry window **4 November 2026**, single day (Wednesday). Harvester legs: **PubMed** health **12**, sensing **0**; **Europe PMC 17** (1 AGRICOLA retro-index record dropped); **Crossref by ISSN 53**; **arXiv** no in-window entry; **OpenAlex** HTTP 429 on the harvester leg. 81 unique after cross-source dedupe, 8 already carried, **73 fresh**; PubMed connector (EDAT 4 Nov, three axes) 22 PMIDs, 21 bucketed to 4 Nov, 12 appended — **85 screened**. Abstracts: 24 recovered by efetch; 20 records, mostly Elsevier day-0 deposits, stayed empty. **Amass** BiomedCore (ingest-date filter) 15 returned: 7 already in `seen.json`, 1 previously rejected, **1 admitted as backfill**, 6 held for a later slot because the backfill cap was reached. **Consensus** calibration sweep: 10 hits, 1 new by DOI (ARMIE, admitted as backfill). The third backfill (Goyal) was held from the previous run's Consensus sweep. **OpenAlex, alphaXiv, bioRxiv/medRxiv and Wiley Scholar Gateway connectors were not available on this session**, so those legs did not run. **Scite** retraction check **unavailable** — the connector again refused with a paid-plan error, so no record here carries a Scite retraction check. **18 admitted** (15 window, 3 backfill), **70 rejected** with logged reasons. **Trial watch** 4–6 November: ClinicalTrials.gov 1 hit, NCT07914010 (REACT), already carried; Amass TrialCore 15 returned, relevant ones already carried — **0 new**. **3 carried on metadata alone**. The x2 funnel is drawn (15 ratio estimates) but Egger is withheld: 15 estimates from 4 sources are not independent. Abstracts remain the copyright of their publishers.